

PRACTICAL TIPS & TRICKS

1. HYPOTHESIS AND SIGNIFICANCE TESTING FOR MEANS & VARIANCES:

- MEAN $\rightarrow$ average of the given set of data (vector)
- VARIANCE $\rightarrow$ sum of squares of differences between all numbers and the mean over the no. of elements in the given set of data
- STANDARD DEVIATION $\rightarrow$ the square root of variance
- $\rightarrow$ mean(X) $\rightarrow$ computes the mean
- $\rightarrow$ var(X) $\rightarrow$ computes the variance
- $\rightarrow$ std(X) $\rightarrow$ computes the standard deviation
- $\rightarrow$ $H_0 \rightarrow$ null hypothesis
- $\rightarrow$ $H_1 \rightarrow$ alternative hypothesis
- $\rightarrow H_0 : \theta = \theta_0 \rightarrow$ we always take the null hypothesis to be the equality
- $\rightarrow H_1 :$
 - $\theta < \theta_0 \Rightarrow$ left-tailed test
 - $\theta > \theta_0 \Rightarrow$ right-tailed test
 - $\theta \neq \theta_0 \Rightarrow$ two-tailed test
- $\rightarrow$ alpha: $\alpha \in (0, 1) \rightarrow$ the significance level
- $\rightarrow$ TS $\rightarrow$ statistics test
- $\rightarrow$ $TS_0 = TS(\theta = \theta_0) \rightarrow$ observed value of the statistic test
- $\rightarrow$ RR $\rightarrow$ rejection region
- $\rightarrow$ P-value $\rightarrow$ probability value
- $\rightarrow$ sigma σ is the VARIANCE

$$\text{var}^2 = \text{std}$$

1.1. ZTEST :

$$\circ [H, P, CI, ZVAL] = \text{ztest}(X, M, SIGMA, ALPHA, TAIL)$$

- X → the vector
- M → testing value (θ_0)
- $SIGMA$ → can be given in the statement or computed ($\text{std}(X)$)
- $ALPHA$ → the significance level (usually $5\% = 0.05$)
- $TAIL = \begin{cases} -1, & \text{for left-tailed test} \\ 0, & \text{for bilateral test} \\ 1, & \text{for right-tailed test} \end{cases}$
- H → tells us if we reject the hypothesis or not ($0 \rightarrow \text{no}$, $1 \rightarrow \text{reject}$)
- P → the P-value of the test
- CI → the confidence interval, containing the lower & upper boundaries of $100(1-\alpha)\%$
- $ZVAL = TS_0$ → the value of the statistics test

◦ rejection region (RR) :

→ for ztest we will use norminv

a) left-tailed test $\theta < \theta_0$:

- * $RR = (-\infty, tt_\alpha)$
- * $tt_\alpha = \text{norminv}(\alpha)$

b) right-tailed test $\theta > \theta_0$:

- * $RR = (tt_{1-\alpha}, \infty)$
- * $tt_{1-\alpha} = \text{norminv}(1 - \alpha)$

c) two-tailed test $\theta \neq \theta_0$:

- * $RR = (-\infty, tt_{\frac{\alpha}{2}}) \cup (tt_{1-\frac{\alpha}{2}}, \infty)$
- * $tt_{\frac{\alpha}{2}} = \text{norminv}(\alpha/2)$
- * $tt_{1-\frac{\alpha}{2}} = \text{norminv}(1 - \alpha/2)$

◦ when we use it :

→ when we have to test the mean of a population and sigma σ is known

1.2. TTEST :

◦ $[H, P, CI, STATS] = \text{ttest}(X, M, ALPHA, TAIL)$

- X → the vector
- M → the testing value (θ_0)
- $ALPHA$ → the significance level (usually $5\% = 0.05$)
- $TAIL = \begin{cases} -1, & \text{for left-tailed test} \\ 0, & \text{for bilateral test} \\ 1, & \text{for right-tailed test} \end{cases}$

→ H → tells us if we reject the null hypothesis or not
 $H = \begin{cases} 0, & \text{do not reject } H_0 \\ 1, & \text{reject } H_0 \end{cases}$

→ P → the P-value of the test

→ CI → the confidence interval

→ $STATS$ → test statistics

$STATS \begin{cases} \rightarrow .tstat = TS_0 \rightarrow \text{value of the statistics test} \\ \rightarrow .df \rightarrow \text{degrees of freedom} \\ \rightarrow .sd \rightarrow \text{estimated population standard deviation} \end{cases}$

◦ rejection region (RR) :

→ for ttest we will use the student distribution : $T(n-1)$, tinv

a) left-tailed test $\theta < \theta_0$:

→ $RR = (-\infty, tt_\alpha)$

→ $tt_\alpha = \text{tinv}(\alpha, n-1)$, where n is the length of the vector X

b) right-tailed test $\theta > \theta_0$:

→ $RR = (tt_{1-\alpha}, \infty)$

→ $tt_{1-\alpha} = \text{tinv}(1-\alpha, n-1)$

b) two-tailed test $\theta \neq \theta_0$:

→ $RR = (-\infty, tt_{\frac{\alpha}{2}}) \cup (tt_{1-\frac{\alpha}{2}}, \infty)$

→ $tt_{\frac{\alpha}{2}} = \text{tinv}(\alpha/2, n-1)$

→ $tt_{1-\frac{\alpha}{2}} = \text{tinv}(1-\alpha/2, n-1)$

◦ when do we use it :

→ when we have to test the mean of a population and sigma σ is unknown

1.3. VARTEST2 :

◦ $[H, P, CI, STATS] = \text{vartest2}(X_1, X_2, \text{ALPHA}, \text{TAIL})$

$\left\{ \begin{array}{l} \rightarrow X_1 \rightarrow \text{the first vector} \\ \rightarrow X_2 \rightarrow \text{the second vector} \\ \rightarrow \text{ALPHA} \rightarrow \text{the significance level (usually } 5\% = 0.05) \\ \rightarrow \text{TAIL} = \begin{cases} -1, & \text{for left-tailed test} \\ 0, & \text{for bilateral test} \\ 1, & \text{for right-tailed test} \end{cases} \end{array} \right.$

$\left\{ \begin{array}{l} \rightarrow H \rightarrow \text{tells us if we reject the null hypothesis or not} \\ H = \begin{cases} 0, & \text{do not reject } H_0 \\ 1, & \text{reject } H_0 \end{cases} \\ \rightarrow P \rightarrow \text{the P-value of the test} \\ \rightarrow CI \rightarrow \text{the confidence interval} \\ \rightarrow \text{STATS} \begin{cases} \rightarrow t_{\text{stat}} = TS_0 \rightarrow \text{the value of statistic test} \\ \rightarrow df_1 \rightarrow \text{numerator degrees of freedom of the test} \\ \rightarrow df_2 \rightarrow \text{denominator degrees of freedom of the test} \end{cases} \end{array} \right.$

◦ Rejection region (RR) :

→ for vartest2 we will use finv

a) left-tailed test $\theta < \theta_0$:

→ $RR = (-\infty, tt_{\alpha})$

→ $tt_{\alpha} = \text{finv}(\text{alpha}, n_1 - 1, n_2 - 1)$ $\begin{cases} n_1 = \text{length of vector } X_1 \\ n_2 = \text{length of vector } X_2 \end{cases}$

b) right-tailed test $\theta > \theta_0$:

→ $RR = (tt_{1-\alpha}, \infty)$

→ $tt_{1-\alpha} = \text{finv}(1 - \text{alpha}, n_1 - 1, n_2 - 1)$

c) two-tailed test $\theta \neq \theta_0$:

→ $RR = (-\infty, tt_{\frac{\alpha}{2}}) \cup (tt_{1-\frac{\alpha}{2}}, \infty)$

→ $tt_{\frac{\alpha}{2}} = \text{finv}(\text{alpha} / 2, n_1 - 1, n_2 - 1)$

→ $tt_{1-\frac{\alpha}{2}} = \text{finv}(1 - \text{alpha} / 2, n_1 - 1, n_2 - 1)$

◦ when we use it :

→ when we have to test the variances of two populations

1.4. TTEST2 :

◦ $[H, P, CI, STATS] = \text{ttest2}(X_1, X_2, \text{ALPHA}, \text{TAIL}, \text{VARTYPE})$ ← omit it

→ X_1 → the first vector
 → X_2 → the second vector
 → ALPHA → the significance level (usually $5\% = 0.05$)
 → $\text{TAIL} = \begin{cases} -1, & \text{for left-tailed test} \\ 0, & \text{for two-tailed test} \\ 1, & \text{for right-tailed test} \end{cases}$

→ H → tells us if we reject the null hypothesis or not
 $H < 0$ → do not reject H_0
 $H > 0$ → reject H_0
 → P → the P-value of the statistic test
 → $\text{STATS} \begin{cases} \text{tstat} = T_{\text{So}} \rightarrow \text{the value of the statistic test} \\ \text{df} \rightarrow \text{degrees of freedom of the test} \\ \text{std} \rightarrow \text{estimation of the standard deviation} \end{cases}$

◦ rejection region (RR) :

→ for ttest2 we will use tinv , but we have more cases :

A. if the populations variances are equal (aka H from varTest2 is 0) :

A. a) left-tailed test $\theta < \theta_0$:

- $RR = (-\infty, t_{t\alpha})$
- $n = n_1 + n_2 - 2$
- $t_{t\alpha} = \text{tinv}(\alpha, n)$

n_1, n_2 → lengths of the vectors

A. b) right-tailed test $\theta > \theta_0$:

- $RR = (t_{1-\alpha}, \infty)$
- $n = n_1 + n_2 - 2$
- $t_{1-\alpha} = \text{tinv}(1-\alpha, n)$

A. c) two-tailed test $\theta \neq \theta_0$:

- $RR = (-\infty, t_{\frac{\alpha}{2}}) \cup (t_{1-\frac{\alpha}{2}}, \infty)$
- $n = n_1 + n_2 - 2$
- $t_{\frac{\alpha}{2}} = \text{tinv}(\alpha/2, n)$
- $t_{1-\frac{\alpha}{2}} = \text{tinv}(1-\alpha/2, n)$

B. if the populations variances are different (aka H_0 from var test 2 is 1):

$$\rightarrow c = \frac{\frac{s_1}{n_1}}{\frac{s_1}{n_1} + \frac{s_2}{n_2}}$$

$$\rightarrow n = \frac{c^2}{n_1 - 1} + \frac{(1-c)^2}{n_2 - 1}$$

$$\rightarrow n = 1/n$$

B. a) left-tailed test $\theta < \theta_0$:

$$\rightarrow RR = (-\infty, tt_\alpha)$$

$$\rightarrow tt_\alpha = \text{tinv}(\alpha, n)$$

B. b) right-tailed test $\theta > \theta_0$:

$$\rightarrow RR = (tt_{1-\alpha}, \infty)$$

$$\rightarrow tt_{1-\alpha} = \text{tinv}(1-\alpha, n)$$

B. c) two-tailed test $\theta \neq \theta_0$:

$$\rightarrow RR = (-\infty, tt_{\frac{\alpha}{2}}) \cup (tt_{1-\frac{\alpha}{2}}, \infty)$$

$$\rightarrow tt_{\frac{\alpha}{2}} = \text{tinv}(\alpha/2, n)$$

$$\rightarrow tt_{1-\frac{\alpha}{2}} = \text{tinv}(1-\alpha/2, n)$$

• when we use it:

→ when we have to test the difference of means of two populations

→ ON AVERAGE is mentioned

2. CONFIDENCE INTERVALS :

2.1. FOR A POPULATION MEAN :

2.1.1. SIGMA KNOWN :

$$\begin{aligned}n &= \text{length}(X); \\g &= \text{norminv}(1 - \alpha/2); \\m &= \text{mean}(X); \end{aligned}$$

$$\text{left} = m - \frac{\sigma}{\sqrt{n}} \cdot g$$

$$\text{right} = m + \frac{\sigma}{\sqrt{n}} \cdot g$$

2.1.2. SIGMA UNKNOWN :

$$\begin{aligned}n &= \text{length}(X); \\g &= \text{tinv}(1 - \alpha/2, n-1); \\m &= \text{mean}(X); \\s &= \text{std}(X); \end{aligned}$$

$$\text{left} = m - \frac{s}{\sqrt{n}} \cdot g$$

$$\text{right} = m + \frac{s}{\sqrt{n}} \cdot g$$

2.2. FOR A POPULATION VARIANCE :

$$\begin{aligned}n &= \text{length}(X); \\g_1 &= \text{chi2inv}(1 - \alpha/2, n-1); \\g_2 &= \text{chi2inv}(\alpha/2, n-1); \\v^2 &= \text{var}(X); \end{aligned}$$

$$\text{left} = \frac{(n-1)v}{g_1}$$

$$\text{right} = \frac{(n-1)v}{g_2}$$

2.3. FOR THE DIFFERENCE OF TWO POPULATION MEANS :

2.3.1. BOTH SIGMA KNOWN :

$$\begin{aligned}n_1 &= \text{length}(X_1); \\n_2 &= \text{length}(X_2); \\g &= \text{norminv}(1 - \alpha/2); \\m_1 &= \text{mean}(X_1); \\m_2 &= \text{mean}(X_2); \end{aligned}$$

$$\text{left} = m_1 - m_2 - g \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$\text{right} = m_1 - m_2 + g \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

2.3.2 SIGMA UNKNOWN AND EQUAL :

$$n_1 = \text{length}(X_1);$$

$$n_2 = \text{length}(X_2);$$

$$n = n_1 + (n_2 - 2);$$

$$m_1 = \text{mean}(X_1);$$

$$m_2 = \text{mean}(X_2);$$

$$v_1 = \text{var}(X_1);$$

$$v_2 = \text{var}(X_2);$$

$$q = \text{tinv}(1 - \alpha/2, n);$$

$$sp = \sqrt{\frac{(n_1 - 1)v_1 + (n_2 - 1)v_2}{n}}$$

$$\text{left} = m_1 - m_2 - q \cdot sp \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$\text{right} = m_1 - m_2 + q \cdot sp \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

2.3.3 SIGMA UNKNOWN AND DIFFERENT :

$$n_1 = \text{length}(X_1);$$

$$n_2 = \text{length}(X_2);$$

$$m_1 = \text{mean}(X_1);$$

$$m_2 = \text{mean}(X_2);$$

$$v_1 = \text{var}(X_1);$$

$$v_2 = \text{var}(X_2);$$

$$c = \frac{\frac{v_1}{n_1}}{\frac{v_1}{n_1} + \frac{v_2}{n_2}}$$

$$n = \frac{c^2}{n_1 - 1} + \frac{(1 - c)^2}{n_2 - 1}$$

$$n = 1/n$$

$$q = \text{tinv}(1 - \alpha/2, n);$$

$$\text{left} = m_1 - m_2 - q \cdot \sqrt{\frac{v_1}{n_1} + \frac{v_2}{n_2}}$$

$$\text{right} = m_1 - m_2 + q \cdot \sqrt{\frac{v_1}{n_1} + \frac{v_2}{n_2}}$$

2.4. FOR THE RATIO OF TWO POPULATION VARIANCES:

$$n_1 = \text{length}(X_1);$$

$$n_2 = \text{length}(X_2);$$

$$v_1 = \text{var}(X_1);$$

$$v_2 = \text{var}(X_2);$$

$$g = \text{fmin}(1 - \alpha/2, n_1 - 1, n_2 - 1)$$

$$g_1 = \text{fmin}(\alpha/2, n_1 - 1, n_2 - 1)$$

$$\text{left} = \frac{1}{g_1} \cdot \frac{v_1}{v_2}$$

$$\text{right} = \frac{1}{g_2} \cdot \frac{v_1}{v_2}$$